

Smartnode AI Call Analysis

An intelligent system that listens to customer calls and automatically sorts them using Machine Learning.

Project Overview

How we built a tool to help Smartnode handle hundreds of support calls every day.

| THE CHALLENGE

The Problem

Customer support teams spend hours listening to phone calls and typing notes manually. It's slow and hard to stay organized.

The Solution

We built a Python pipeline that automatically transcribes voice calls to text and classifies them into helpful categories.

| SORTING THE CALLS



Urgent

Safety issues or emergencies, like broken smart locks.



Open

New bugs or feature requests that need an answer.



Closed

Fixed problems where the customer is happy.

| STEP 1: TEXT-TO-SPEECH

We used **Google Text-to-Speech (gTTS)** to create fake voice samples for our tests.

- Takes call scripts as text.
- Generates clear .mp3 audio files.
- Allows us to test our AI without real recorded calls.



| STEP 2: AI LISTENING

We use **OpenAI Whisper** to "listen" to the audio.

Whisper is a state-of-the-art AI model that converts speech into highly accurate written text.

Result: A clean transcript that the computer can read and understand.

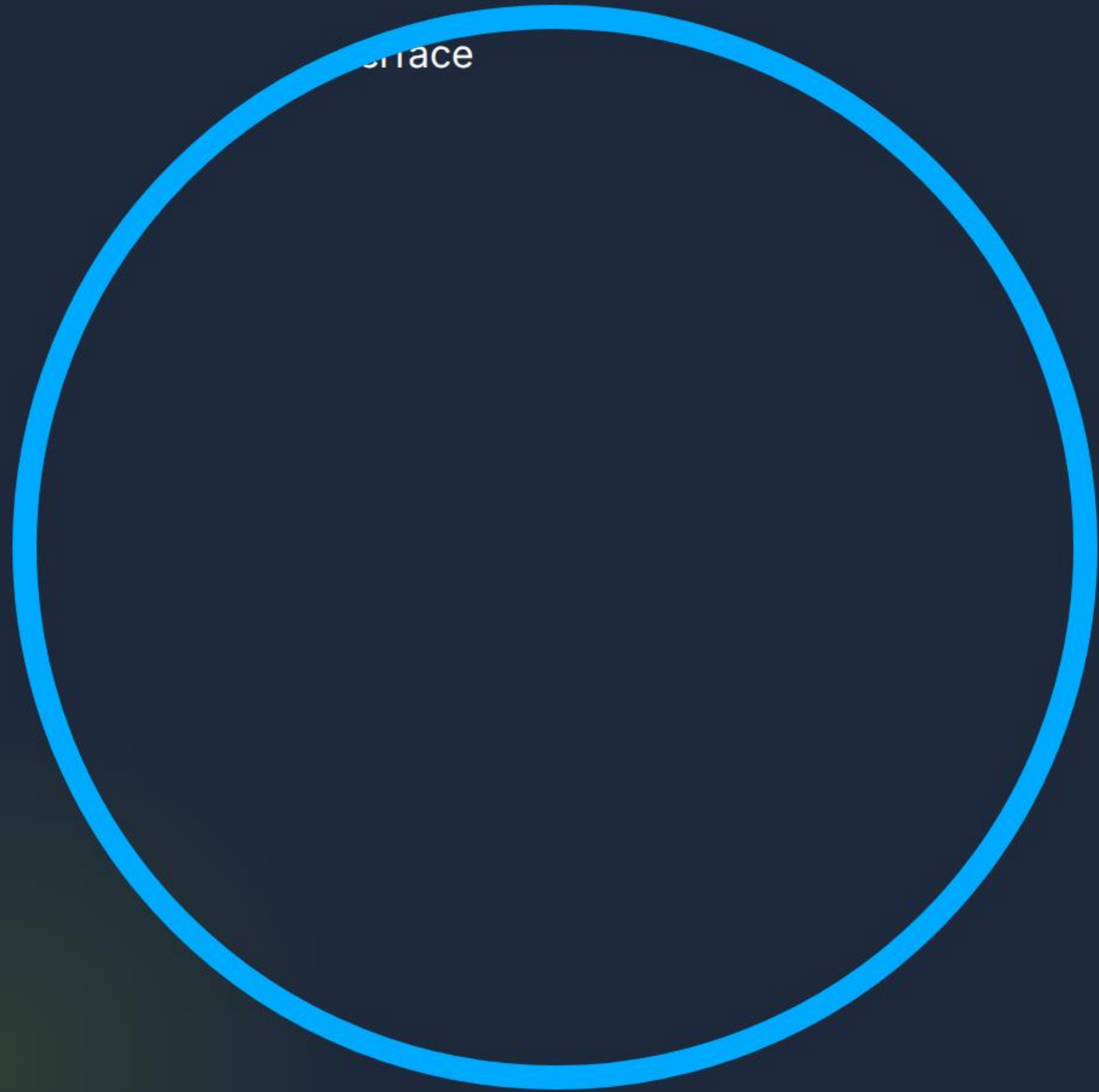


| STEP 3: KEYWORD RULES

- 🔥 **Urgent Keywords:** "Emergency", "Stuck", "Escalate", "System is down".
- ✓ **Closed Keywords:** "Resolved", "Working perfectly", "Thank you".
- ⌚ **Open Keywords:** "Waiting", "Pending", "Update", "Timeline".

The code counts these words to make a quick decision on what the call is about.

STEP 4: SMART AI GUESSING



Advanced Thinking

What if a customer doesn't use the specific keywords? We use **Zero-Shot Classification** (the BART model).

This AI understands the *meaning* of the whole sentence. It can feel the urgency even without the word "Emergency."

CLASSIFICATION ACCURACY

Rule-Based

100% Correct

Smart AI (Zero-Shot)

83% Correct

Rule-based was perfect for our test samples, while the AI proved it's highly capable of understanding general context.

SAMPLE CALL RESULTS

Call ID	Status	Problem Area	AI Prediction
CALL-001	Closed	Wi-Fi Module	Closed (Perfect)
CALL-003	Urgent	Door Lock Stuck	Urgent (Perfect)
CALL-005	Urgent	Hotel System Down	Urgent (Perfect)
CALL-006	Open	App Scene Bug	Open (Perfect)

| THE DATA JOURNEY

